$T: V \to W$

$\ker(T) = \{ v \in V : T(v) = 0 \} \Rightarrow Av = 0$

$\text{Range}(T) = \{ T(v) : v \in V \}$.

$\downarrow$

$\text{Image}(T)$

$T: \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} x+0y+0z \\ 0x+1y+0z \\ 0x+0y+0z \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{Kernel}(T) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x = 0,\ y = 0,\ z \in \mathbb{R} \right\}$$

$$Ax = 0 \qquad \begin{pmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{pmatrix}$$

$x = 0,\ y = 0$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} = z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Image / Range of T

$$Col(A) = \text{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$Range(T) = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} : z = 0 \right\}$$

$T: \mathbb{R}^4 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} x-y+z+t \\ x+2z-t \\ x+y+3z-3t \end{pmatrix}$$

Find the Kernel(T) & Range/Image(T)

Kernel (T)

$$A = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 0 & 2 & -1 \\ 1 & 1 & 3 & -3 \end{pmatrix}$$

$AX = 0$

$$\begin{pmatrix} 1 & -1 & 1 & 1 & 0 \\ 1 & 0 & 2 & -1 & 0 \\ 1 & 1 & 3 & -3 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & -1 & 1 & 1 & 0 \\ 0 & 1 & 1 & -2 & 0 \\ 0 & 2 & 2 & -4 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & -1 & 1 & 1 & 0 \\ 0 & 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$y + z - 2t = 0$

$\boxed{y = -z + 2t}$

$x - (-z + 2t) + z + t = 0$

$x + 2z - t = 0$

$x = -2z + t$

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} -2z + t \\ -z + 2t \\ z \\ t \end{pmatrix}$$

$$= z\begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

Kernel(T) = Span{ , }

$$A=\begin{pmatrix}1 & -1 & 1 & 1\\ 1 & 0 & 2 & -1\\ 1 & 1 & 3 & -3\end{pmatrix}$$

Span of columns of A.

$$c_1\begin{pmatrix}1\\1\\1\end{pmatrix}+c_2\begin{pmatrix}-1\\0\\1\end{pmatrix}+c_3\begin{pmatrix}1\\2\\3\end{pmatrix}+c_4\begin{pmatrix}1\\-1\\-3\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\left(\begin{array}{cccc|c}1 & -1 & 1 & 1 & x\\ 1 & 0 & 2 & -1 & y\\ 1 & 1 & 3 & -3 & z\end{array}\right)$$

$$\underset{R}{\sim}\left(\begin{array}{cccc|c}1 & -1 & 1 & 1 & x\\ 0 & 1 & 1 & -2 & -x+y\\ 0 & 2 & 2 & -4 & -x+z\end{array}\right)\underset{R}{\sim}\left(\begin{array}{cccc|c}1 & -1 & 1 & 1 & x\\ 0 & 1 & 1 & -2 & -x+y\\ 0 & 0 & 0 & 0 & 2x-2y-x+z\end{array}\right)$$

$$x-2y+z=0$$

Image(T) = Range(T)

$$=\left\{\begin{pmatrix}x\\y\\z\end{pmatrix}: x-2y+z=0\right\}.$$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}x\\y\\-x+2y\end{pmatrix}=x\begin{pmatrix}1\\0\\-1\end{pmatrix}+y\begin{pmatrix}0\\1\\2\end{pmatrix}$$

Find the kernel & Range of D

$D: P_3 \to P_2$

$D(p(x)) = p'(x)$

$B = \{1, x, x^2, x^3\}$, $C = \{1, x, x^2\}$

$D(1) = 0$ ✓

$D(x) = 1$ ✓

$D(x^2) = 2x$ ✓

$D(x^3) = 3x^2$ ✓

$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$

$0 + 2x + 0x^2$ $\quad$ $0 + 0x + 3x^2$

Kernel

$Ax = 0$

$\left(\begin{array}{cccc|c} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \end{array}\right)$

$x = 0$ $\quad 2y = 0 \Rightarrow y = 0$

$3z = 0 \Rightarrow z = 0$

$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} w \\ 0 \\ 0 \\ 0 \end{pmatrix} = w \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$\text{Kernel}(T) = \{ax^3 + bx^2 + cx + d : b = c = d = 0\}$

Range / Image

$\left(\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{array}\right)\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$

$\text{Range}(T) = \{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$

Kernel & Range/Image

$T: V \to W$

Step-I: Matrix of transformation.

Step-II: Kernel $AX = O \Rightarrow X = ?$

Kernel.

Step-III: Range / Image Columnspace (A) $\Rightarrow$ Span of columns of A.